

Point 1 briefing — Forward validation (Step 0)

Does the shallow-water formula recover HydroLight Rrs given true depth and true bottom reflectance?

Shared setup: structure and coefficients of the four formulas

Lee et al. (1998, 1999) give a semi-analytical shallow-water model in two structural forms: a single attenuation factor X , and a two-factor form with separate water-column and bottom terms. Equations ① and ② use the single- X form; Equations ③ and ④ use the two- X form. Each structure is run once with the original Lee coefficients and once with coefficients fitted on an independent set of 972 radiative-transfer simulations. The four combinations are the formulas validated here (Table C-1). Viewing is nadir, so $\cos(\theta_v^w) = 1$. The optical ratios are $u = b_b / (a + b_b)$ and $\kappa = a + b_b$.

Table C-1. The four formulas used in the forward validation.

Eq.	Structure	Du coefficients	Source
①	single- X	$DuC = 1.03 \sqrt{1 + 2.4u}$ $XC = XB$	Lee et al. (1998, 1999)
②	single- X	$DuC = C1 \sqrt{1 + C2 \cdot u}$ $C1 = 1.64670, C2 = 0.03234$	Fitted on 972 independent simulations
③	two- X	$DuC = 1.03 \sqrt{1 + 2.4u}$ $DuB = 1.04 \sqrt{1 + 5.4u}$	Lee et al. (1998, 1999)
④	two- X	$DuC = C1 \sqrt{1 + C2 \cdot u}, C1 = 0.50, C2 = 8$ $DuB = B1 \sqrt{1 + B2 \cdot u} = 1.61277 (B2 = 0)$	Fitted on 972 independent simulations

Table C-2. HydroLight conditions used to fit the empirical coefficients of Eqs. ② and ④ ($n = 972$). This set does not overlap the 288 scenes of Point 1.

Item	Levels
Water type	Classic Case-2
Solar zenith angle	10°, 30°, 50°
Chlorophyll-a	0.1, 0.3, 0.5 mg m ⁻³
TSM	0.1, 0.3, 0.5 mg m ⁻³
Bottom / depth / bands	Same family as Table 1; 972 cases in total
Independence from Point 1	Point 1 uses 20°/40°, Chl-a 0.2/0.4 mg m ⁻³ , TSM 0.2/0.4 mg m ⁻³ — no overlap

1. Forward validation — does the formula recover HydroLight Rrs?

Figure 1. Advisor comment (English rendering)

Step 0 — prior validation of the forward model. Depth inversion and bottom unmixing should not begin until the semi-analytical model is shown to be consistent with observed shallow-water Rrs. If known depth H and bottom reflectance $\rho(\lambda)$ are inserted and the modelled Rrs still fails to recover the observed spectrum, later errors cannot be attributed to IOP estimation, the attenuation coefficients, or the H - ρ trade-off. Instruction: using Lee et al.'s shallow-water forward model with known H and ρ , first confirm that the modelled Rrs reproduces both the magnitude and the shape of the observed shallow-water spectrum.

Problem

The long-term goal is to invert observed remote-sensing reflectance Rrs for water depth and bottom type. A mismatch with HydroLight shallow-water Rrs cannot, from inversion scores alone, be assigned to the shallow-water formula, to errors in the inherent optical properties (a , bb), or to a trade-off between depth and bottom reflectance. Earlier reports placed a forward calculation that used true depth in a retrieval-performance section; that layout could be misread as a depth-retrieval score, and it did not separate error by IOP source.

Purpose of this step

This step does not retrieve depth. True depth and true bottom reflectance are inserted into the shallow-water formula. The question is whether modelled Rrs reproduces HydroLight shallow-water Rrs in both magnitude and spectral shape under identical conditions. Holding depth and bottom at truth and changing only the source of a and bb partitions the residual into formula/Du error versus IOP-estimation error. That partition is required before later inversion error can be interpreted.

Procedure

For each scene, only true depth and true bottom reflectance are inserted. The four formulas (①–④) compute modelled Rrs at 443, 490, 560 and 665 nm, which is then compared with HydroLight shallow-water Rrs. Case A uses HydroLight truth for a and bb . Case B uses a and bb obtained by applying QAA to infinite-depth Rrs of the same water. Depth inversion and bottom mixing are not performed.

The validation set comprises 288 HydroLight radiative-transfer simulations of Classic Case-2 water: 6 bottom types $\times$ 6 depths $\times$ 2 solar zenith angles $\times$ 2 chlorophyll- a levels $\times$ 2 TSM levels (Table 1). Quantities inserted into the formula, and the comparison target, are listed in Table 2. Empirical coefficients of Eqs. ② and ④ were estimated under conditions independent of these 288 scenes and are not re-estimated here.

Metrics are RMSE and mean bias of the Rrs residual (model – HydroLight). Overall scores use 288 scenes $\times$ 4 bands = 1,152 points; band-wise scores use 288 points. In addition, to test whether the formula's DuC and DuB match HydroLight, Du inverted from HydroLight Rrs with true depth and bottom ("HydroLight-inverted Du") is compared with the formula's Du (Appendix A).

Table 1. HydroLight radiative-transfer conditions. The factorial design is $6 \times 6 \times 2 \times 2 \times 2 = 288$ cases.

Item	Setting
Water type	Classic Case-2
Bottom type	ooid sand, dark sediment, coral, brown algae, red algae, green algae (6)
Depth	0.5, 1, 2, 3, 4, 5 m
Solar zenith angle	20°, 40°
Wavelength	443, 490, 560, 665 nm
Chlorophyll-a	0.2, 0.4 mg m ⁻³
TSM	0.2, 0.4 mg m ⁻³
Temperature / salinity / sky	20 °C / 35 PSU / clear sky (cloud = 0)

Table 2. Experimental design of the forward validation. Depth is not retrieved in either case.

Case	Inserted into the formula	Compared against	Formulas	Data
A	True depth, true bottom ρ , HydroLight a and bb	HydroLight shallow Rrs	①–④	288
B	True depth, true bottom ρ , QAA a and bb	HydroLight shallow Rrs	①–④	288

Results

With true depth and true bottom reflectance prescribed, all four formulas recover the spectral shape of HydroLight shallow-water Rrs (Figures 2–9). Magnitude error (overall RMSE, Table 3) in Case A ranks as ② 1.49×10^{-3} , ④ 1.52×10^{-3} , ③ 1.68×10^{-3} , ① 2.44×10^{-3} . Eqs. ② and ④ (empirical coefficients) are smaller than Eqs. ① and ③ (original Lee coefficients). These numbers are not depth-retrieval scores.

When the IOPs are taken from QAA (Case B), RMSE of ① and ③ rises to 3.11×10^{-3} and 2.51×10^{-3} . Under the same switch, ② and ④ become 1.36×10^{-3} and 1.45×10^{-3} , similar to or slightly smaller than Case A. The extra error in ① and ③ is therefore due mainly to IOP estimation, whereas ② and ④ already have a smaller formula error and are little affected by the QAA switch. Mean bias is positive for ① and ③ (the model is brighter than HydroLight) and near zero or weakly negative for ② and ④. The Du comparison (Appendix A) shows that original Lee Du disagrees with the HydroLight-inverted value: in the single-X form (①), DuC is smaller than the inverted value; in the two-X form (③), the water-column term DuC is too large and the bottom term DuB is too small. Either mismatch brightens modelled Rrs, confirming that the positive bias of ① and ③ originates in the Du coefficients.

Eq.	Structure / coefficients	Case A RMSE	Case A Bias	Case B RMSE	Case B Bias	Interpretation
①	single-X, Lee	2.44	+1.37	3.11	+1.68	Rises in Case B. Extra error is IOP
②	single-X, empirical	1.49	-0.10	1.36	+0.18	Smallest. Insensitive to the QAA switch
③	two-X, Lee	1.68	+0.78	2.51	+1.08	Rises in Case B. Extra error is IOP
④	two-X, empirical	1.52	-0.68	1.45	-0.40	Similar to ②. Insensitive to the QAA switch

Table 3. Overall scores ($\times 10^{-3}$; 288 scenes $\times$ 4 bands = 1,152 points). RMSE = sqrt of the mean squared residual (model Rrs – HydroLight Rrs); Bias = mean(model – HydroLight). Positive bias means the model is brighter than HydroLight. Orange background = Case B. These are not depth-retrieval scores.

Conclusion

The Step 0 question is this: with true depth and true bottom reflectance inserted, does modelled R_{rs} recover both the shape and the magnitude of HydroLight shallow-water R_{rs} ?

Shape is recovered by all four formulas (Figures 2–9: the 560 nm peak and the 665 nm drop match HydroLight). Magnitude is judged on Case A (HydroLight-true a , b_b). Overall RMSE and bias ($\times 10^{-3}$, Table 3, 1,152 points) are ② 1.49 (bias -0.10), ④ 1.52 (-0.68), ③ 1.68 (+0.78), ① 2.44 (+1.37). Only Eq. ② has near-zero bias and the smallest RMSE, so both shape and magnitude match. Eqs. ① and ③ match the shape but are biased bright, so magnitude does not match; the cause is original Lee Du disagreeing with the HydroLight-inverted value (Appendix A).

The formula to adopt is Eq. ② (single-X, empirical coefficients). Use Eq. ④ if a two-X structure is required. Do not use Eqs. ① or ③ as the working formula.

Later depth and bottom inversion proceeds with Eq. ②. The residual that remains in Case A is formula/Du. When IOPs are taken from QAA (Case B), RMSE rises only for ① (2.44→3.11) and ③ (1.68→2.51), while ② stays essentially unchanged (1.49→1.36). Later error is therefore partitioned into formula error versus QAA error.

A. Spectral shape — modelled R_{rs} vs HydroLight R_{rs} (one formula, one case per panel)

The left column is Case A and the right column is Case B. Each panel pools 288 simulations by wavelength and plots the mean and the median. Where blue (model) and red (HydroLight) overlap, shape and magnitude agree. Depth is not retrieved.

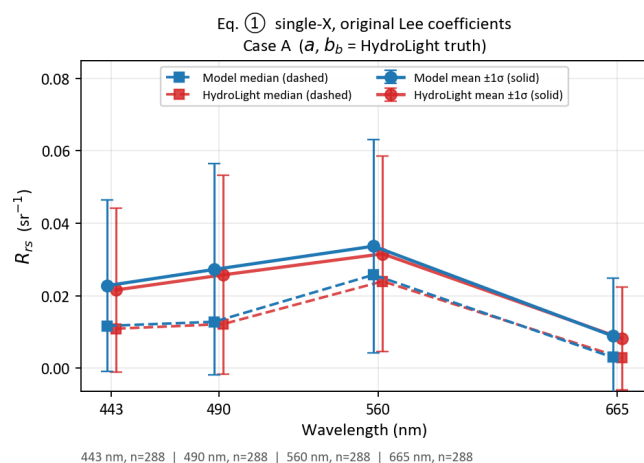


Figure 2. Eq. ① single-X, original Lee coefficients, Case A (a , b_b = HydroLight truth). Wavelength (nm) on the x-axis; R_{rs} (sr^{-1}) on the y-axis. Blue = model; red = HydroLight. Solid line with circles = mean of 288 scenes; vertical bars = $\pm 1\sigma$; dashed line with squares = median. The two traces are offset by ± 2 nm so they do not overplot.

Reading (Eq. ① (single-X, Lee)). Red and blue follow the same shape (a peak at 560 nm and a drop at 665 nm), but blue (model) lies above red. Mean bias is $+1.37 \times 10^{-3}$ (A) $\rightarrow +1.68 \times 10^{-3}$ (B), so the gap widens in Case B. The original-Lee single-X formula is biased bright.

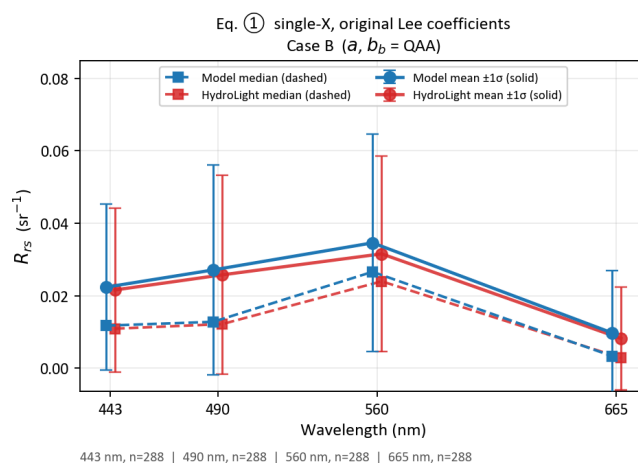


Figure 3. Eq. ① single-X, original Lee coefficients, Case B (a , b_b = QAA). Wavelength (nm) on the x-axis; R_{rs} (sr^{-1}) on the y-axis. Blue = model; red = HydroLight. Solid line with circles = mean of 288 scenes; vertical bars = $\pm 1\sigma$; dashed line with squares = median. The two traces are offset by ± 2 nm so they do not overplot.

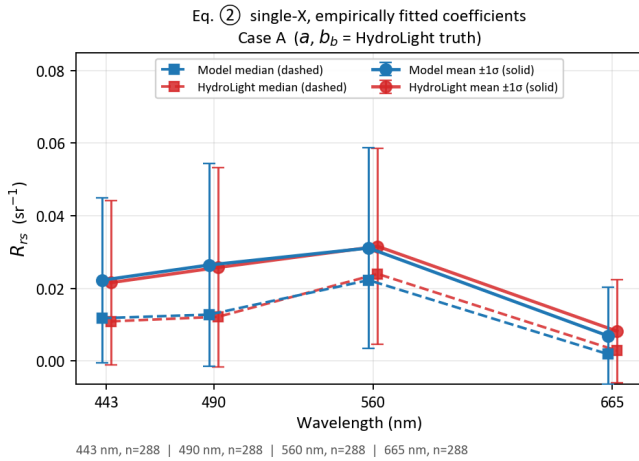


Figure 4. Eq. ② single-X, empirically fitted coefficients, Case A ($a, b_b = \text{HydroLight truth}$). Wavelength (nm) on the x-axis; R_{rs} (sr^{-1}) on the y-axis. Blue = model; red = HydroLight. Solid line with circles = mean of 288 scenes; vertical bars = $\pm 1\sigma$; dashed line with squares = median. The two traces are offset by ± 2 nm so they do not overlap.

Reading (Eq. ② (single-X, empirical)). In both Case A and Case B, red and blue nearly overlap in mean, median and $\pm 1\sigma$ width. Mean bias is -0.10×10^{-3} (A) and $+0.18 \times 10^{-3}$ (B). This is the best spectral reproduction of the four formulas.

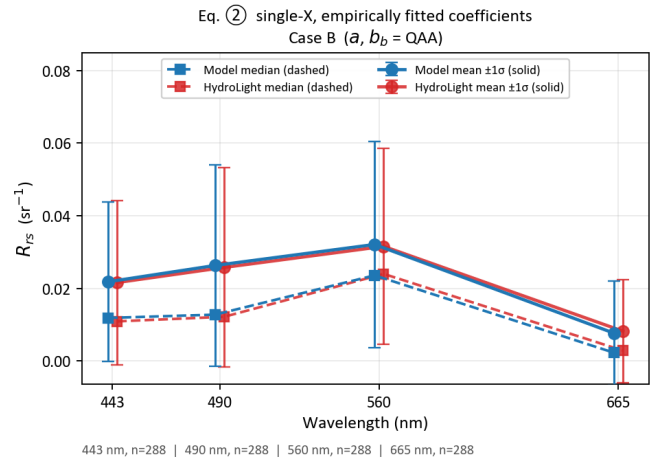


Figure 5. Eq. ② single-X, empirically fitted coefficients, Case B ($a, b_b = \text{QAA}$). Wavelength (nm) on the x-axis; R_{rs} (sr^{-1}) on the y-axis. Blue = model; red = HydroLight. Solid line with circles = mean of 288 scenes; vertical bars = $\pm 1\sigma$; dashed line with squares = median. The two traces are offset by ± 2 nm so they do not overlap.

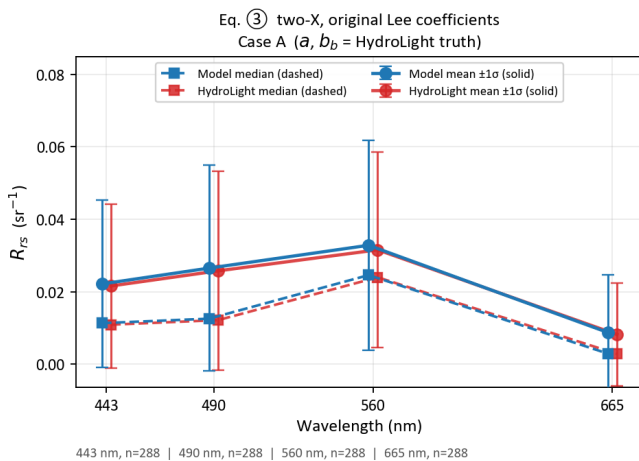


Figure 6. Eq. ③ two-X, original Lee coefficients, Case A ($a, b_b = \text{HydroLight truth}$). Wavelength (nm) on the x-axis; R_{rs} (sr^{-1}) on the y-axis. Blue = model; red = HydroLight. Solid line with circles = mean of 288 scenes; vertical bars = $\pm 1\sigma$; dashed line with squares = median. The two traces are offset by ± 2 nm so they do not overlap.

Reading (Eq. ③ (two-X, Lee)). The shape is correct, but blue lies above red, and the 560–665 nm gap grows in Case B as in Eq. ①. Mean bias is $+0.78 \times 10^{-3}$ (A) $\rightarrow +1.08 \times 10^{-3}$ (B). Splitting X into two terms does not by itself stop QAA IOP error.

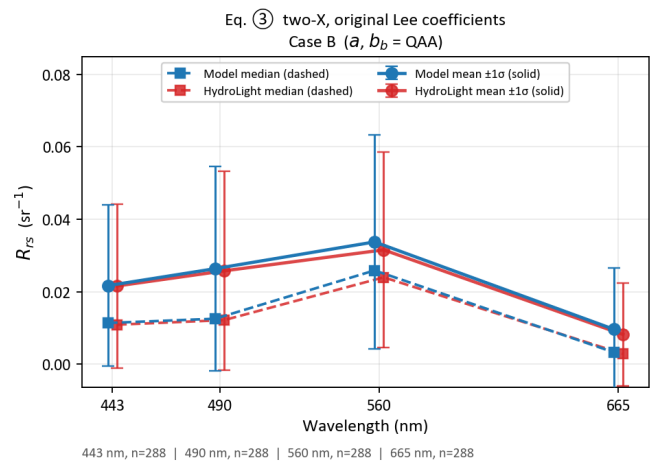


Figure 7. Eq. ③ two-X, original Lee coefficients, Case B ($a, b_b = \text{QAA}$). Wavelength (nm) on the x-axis; R_{rs} (sr^{-1}) on the y-axis. Blue = model; red = HydroLight. Solid line with circles = mean of 288 scenes; vertical bars = $\pm 1\sigma$; dashed line with squares = median. The two traces are offset by ± 2 nm so they do not overlap.

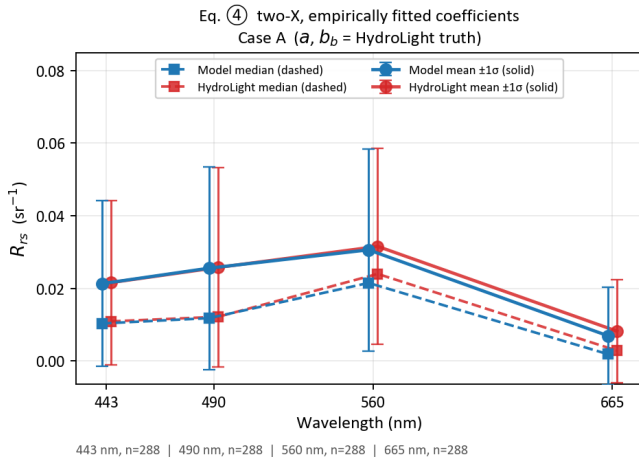


Figure 8. Eq. ④ two-X, empirically fitted coefficients, Case A ($a, b_b = \text{HydroLight truth}$). Wavelength (nm) on the x-axis; R_{rs} (sr^{-1}) on the y-axis. Blue = model; red = HydroLight. Solid line with circles = mean of 288 scenes; vertical bars = $\pm 1\sigma$; dashed line with squares = median. The two traces are offset by ± 2 nm so they do not overplot.

Reading (Eq. ④ (two-X, empirical)). The two colours overlap much as in Eq. ②. Blue lies slightly below red (negative bias, -0.68×10^{-3} in A, -0.40×10^{-3} in B). The shape is preserved under the Case B switch.

B. Magnitude — band-wise R_{rs} RMSE (one formula, one case per panel)

Each bar is RMSE at one wavelength. The whisker is the standard deviation σ of the residual (model – HydroLight) and shows how much the error varies across scenes. The y-axis range is auto-scaled per panel, so compare magnitudes using the numbers on the bars ($\times 10^3$) and Table 4.

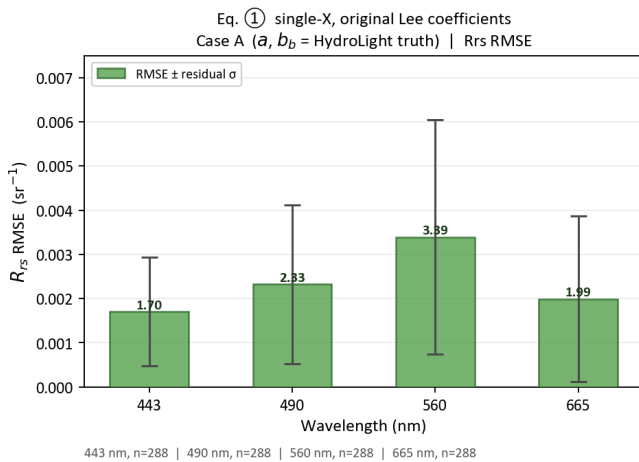


Figure 10. Band-wise R_{rs} RMSE for Eq. ① single-X, original Lee coefficients, Case A ($a, b_b = \text{HydroLight truth}$). Wavelength (nm) on the x-axis; RMSE (sr^{-1}) on the y-axis. Green bars = RMSE (numbers on bars = $\times 10^3$); black whiskers = residual standard deviation $\pm \sigma$. $n = 288$ per band. The y-axis range differs by panel.

Reading (Eq. ① (single-X, Lee)). In Case A, 560 nm is largest (3.39×10^{-3}). In Case B, 560 and 665 nm rise further to 4.65 and 3.43×10^{-3} , while 443 nm decreases. The taller bars are the effect of QAA a, b_b .

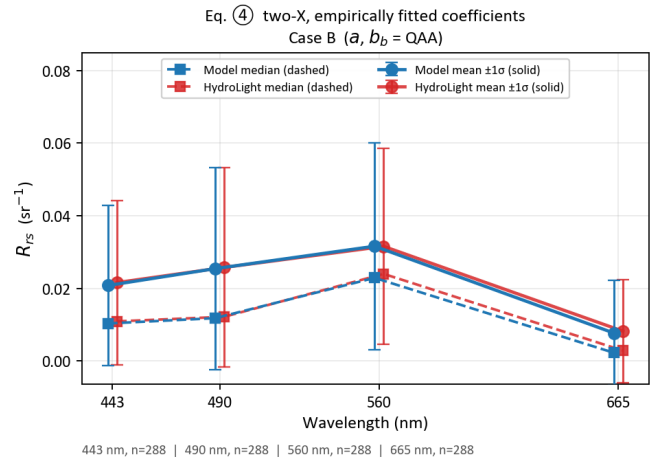


Figure 9. Eq. ④ two-X, empirically fitted coefficients, Case B ($a, b_b = \text{QAA}$). Wavelength (nm) on the x-axis; R_{rs} (sr^{-1}) on the y-axis. Blue = model; red = HydroLight. Solid line with circles = mean of 288 scenes; vertical bars = $\pm 1\sigma$; dashed line with squares = median. The two traces are offset by ± 2 nm so they do not overplot.

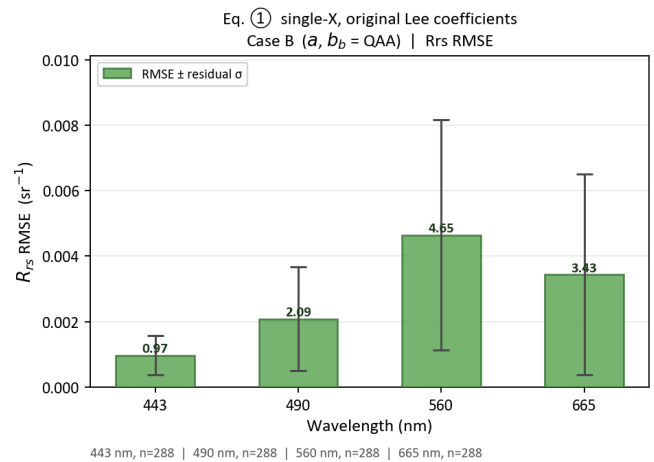


Figure 11. Band-wise R_{rs} RMSE for Eq. ① single-X, original Lee coefficients, Case B ($a, b_b = \text{QAA}$). Wavelength (nm) on the x-axis; RMSE (sr^{-1}) on the y-axis. Green bars = RMSE (numbers on bars = $\times 10^3$); black whiskers = residual standard deviation $\pm \sigma$. $n = 288$ per band. The y-axis range differs by panel.

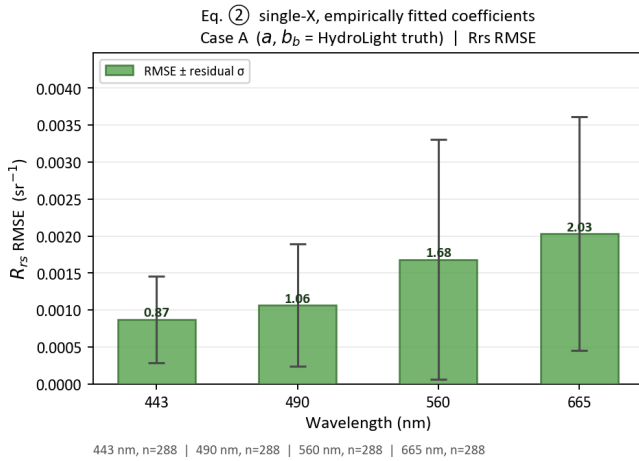


Figure 12. Band-wise Rrs RMSE for Eq. ② single-X, empirically fitted coefficients, Case A (a , b_b = HydroLight truth). Wavelength (nm) on the x-axis; RMSE (sr^{-1}) on the y-axis. Green bars = RMSE (numbers on bars = $\times 10^3$); black whiskers = residual standard deviation $\pm \sigma$. $n = 288$ per band. The y-axis range differs by panel.

Reading (Eq. ② (single-X, empirical)). All four bands are similar in Case A and Case B ($0.9\text{--}2.0 \times 10^{-3}$). At 665 nm, Case B is actually smaller ($2.03 \rightarrow 1.15 \times 10^{-3}$). Formula error is already small, so the QAA switch barely moves the result.

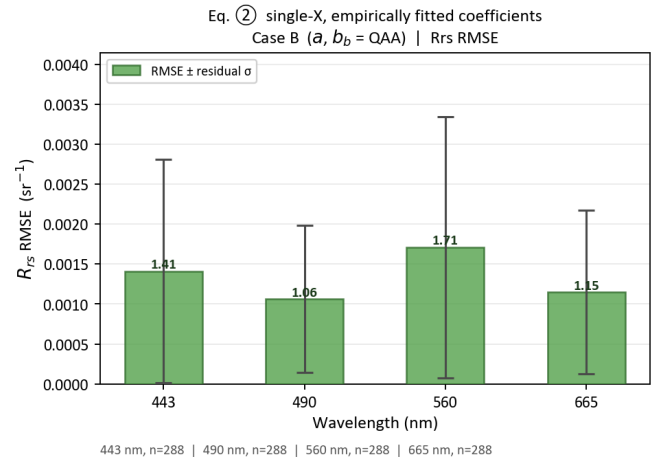


Figure 13. Band-wise Rrs RMSE for Eq. ② single-X, empirically fitted coefficients, Case B (a , b_b = QAA). Wavelength (nm) on the x-axis; RMSE (sr^{-1}) on the y-axis. Green bars = RMSE (numbers on bars = $\times 10^3$); black whiskers = residual standard deviation $\pm \sigma$. $n = 288$ per band. The y-axis range differs by panel.

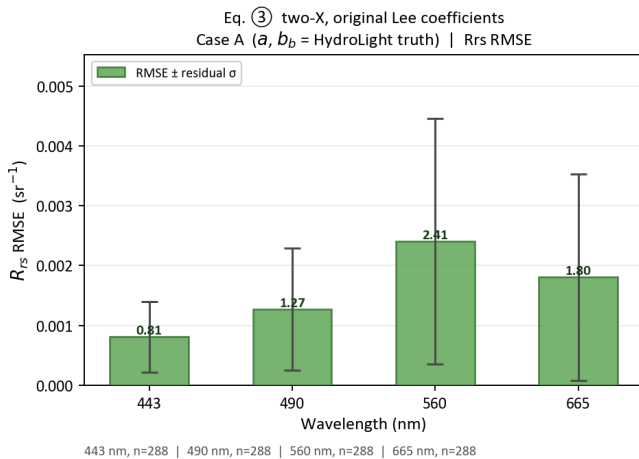


Figure 14. Band-wise Rrs RMSE for Eq. ③ two-X, original Lee coefficients, Case A (a , b_b = HydroLight truth). Wavelength (nm) on the x-axis; RMSE (sr^{-1}) on the y-axis. Green bars = RMSE (numbers on bars = $\times 10^3$); black whiskers = residual standard deviation $\pm \sigma$. $n = 288$ per band. The y-axis range differs by panel.

Reading (Eq. ③ (two-X, Lee)). Case A is lower than Eq. ① (2.41×10^{-3} at 560 nm), but Case B again raises 560 and 665 nm to 3.55 and 3.23×10^{-3} . A two-X structure remains sensitive to IOP error if the original Lee coefficients are used.

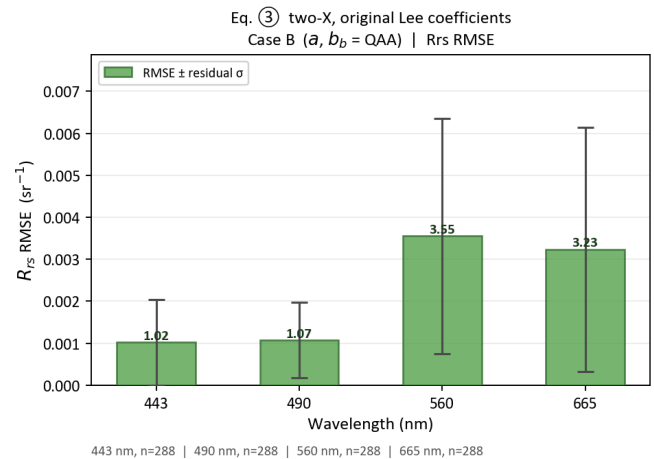


Figure 15. Band-wise Rrs RMSE for Eq. ③ two-X, original Lee coefficients, Case B (a , b_b = QAA). Wavelength (nm) on the x-axis; RMSE (sr^{-1}) on the y-axis. Green bars = RMSE (numbers on bars = $\times 10^3$); black whiskers = residual standard deviation $\pm \sigma$. $n = 288$ per band. The y-axis range differs by panel.

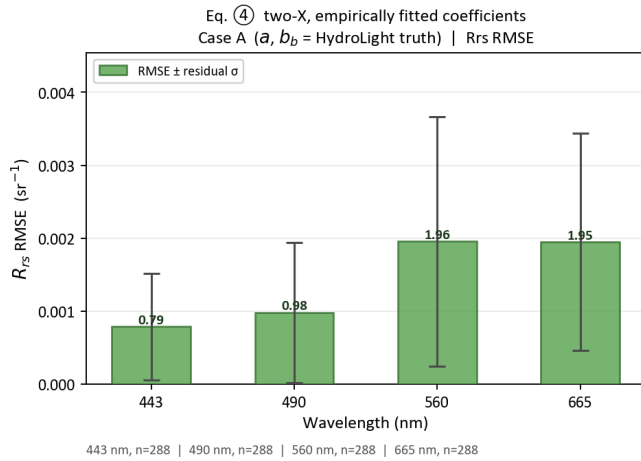


Figure 16. Band-wise Rrs RMSE for Eq. ④ two-X, empirically fitted coefficients, Case A (a, b_b = HydroLight truth). Wavelength (nm) on the x-axis; RMSE (sr^{-1}) on the y-axis. Green bars = RMSE (numbers on bars = $\times 10^3$); black whiskers = residual standard deviation $\pm \sigma$. $n = 288$ per band. The y-axis range differs by panel.

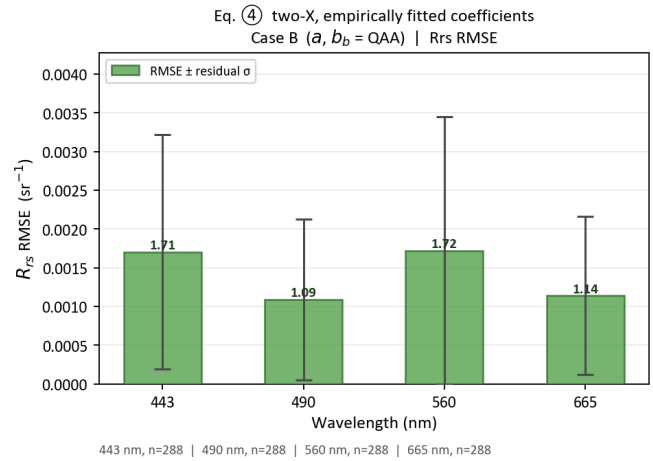


Figure 17. Band-wise Rrs RMSE for Eq. ④ two-X, empirically fitted coefficients, Case B (a, b_b = QAA). Wavelength (nm) on the x-axis; RMSE (sr^{-1}) on the y-axis. Green bars = RMSE (numbers on bars = $\times 10^3$); black whiskers = residual standard deviation $\pm \sigma$. $n = 288$ per band. The y-axis range differs by panel.

Reading (Eq. ④ (two-X, empirical)). The level matches Eq. ②. In Case B only 443 nm increases ($0.79 \rightarrow 1.71 \times 10^{-3}$) while 665 nm decreases ($1.95 \rightarrow 1.14 \times 10^{-3}$). Overall, the formula is insensitive to the QAA switch.

Eq.	Case A (HydroLight a, b_b)				Case B (QAA a, b_b)			
($\times 10^{-3}$)	443	490	560	665	443	490	560	665
①	1.70	2.33	3.39	1.99	0.97	2.09	4.65	3.43
②	0.87	1.06	1.68	2.03	1.41	1.06	1.71	1.15
③	0.81	1.27	2.41	1.80	1.02	1.07	3.55	3.23
④	0.79	0.98	1.96	1.95	1.71	1.09	1.72	1.14

Table 4. Band-wise Rrs RMSE ($\times 10^{-3}$). Matches the bar heights in Figures 10–17. $n = 288$ per cell. Orange background = Case B.

Eq.	Case A (HydroLight a, b_b)				Case B (QAA a, b_b)			
($\times 10^{-3}$)	443	490	560	665	443	490	560	665
①	+1.19	+1.48	+2.12	+0.68	+0.77	+1.35	+3.04	+1.56
②	+0.65	+0.67	-0.45	-1.28	+0.21	+0.54	+0.52	-0.53
③	+0.55	+0.75	+1.26	+0.55	+0.09	+0.60	+2.19	+1.43
④	-0.30	-0.20	-0.96	-1.26	-0.80	-0.33	+0.04	-0.51

Table 5. Band-wise mean bias ($\times 10^{-3}$) = mean(model – HydroLight). Positive = the model is brighter. ① and ③ are positive at all bands (larger at 560 and 665 nm in Case B). ② and ④ have small absolute values and are negative at 665 nm.

Eq.	443 nm	490 nm	560 nm	665 nm	Overall
①	-0.73	-0.24	+1.25	+1.44	+0.67
②	+0.54	+0.00	+0.03	-0.88	-0.13
③	+0.22	-0.20	+1.15	+1.43	+0.83
④	+0.92	+0.11	-0.23	-0.80	-0.07

Table 6. RMSE(Case B) – RMSE(Case A) ($\times 10^{-3}$) = right minus left in Table 4. Positive = error increases when a and b_b are taken from QAA. Depth and bottom are identical in the two cases, so this difference is solely the effect of IOP source. ① and ③ are +1.1 to +1.4 at 560 and 665 nm; ② and ④ are smaller and negative at 665 nm.

Summary. Shape is recovered by all four formulas; magnitude matches only for Eq. ② (Table 3). The formula to adopt is ②. The bright bias of ① and ③ originates in the Du coefficients (Appendix A).

Appendix A. Du comparison — formula DuC, DuB versus HydroLight-inverted Du

Tables 3–6 and Figures 2–17 already support the main-text conclusion. This appendix is supplementary evidence that the Rrs residual is rooted in the Du coefficients.

HydroLight-inverted Du is obtained by inserting HydroLight Rrs, true depth, true bottom reflectance and the case's a and b_b , then solving the formula for X , and setting $Du = X/\kappa - 1/\cos(\theta_w)$. It is the Du that would be required to match HydroLight exactly. Single- X formulas (①, ②) have $Du_C = Du_B$, so only Du_C is plotted. Two- X formulas (③, ④) plot the water-column term Du_C and the bottom term Du_B separately. Inversion fails when the argument of the logarithm leaves $(0, 1)$, so band-wise n can be smaller than 288 (shown under each panel); bands with $n < 30$ are omitted. Inverted values have a long tail, so a few scenes pull the mean; spectral plots in this appendix therefore use the median and the 25–75% interval.

DuC (water-column term) — spectra

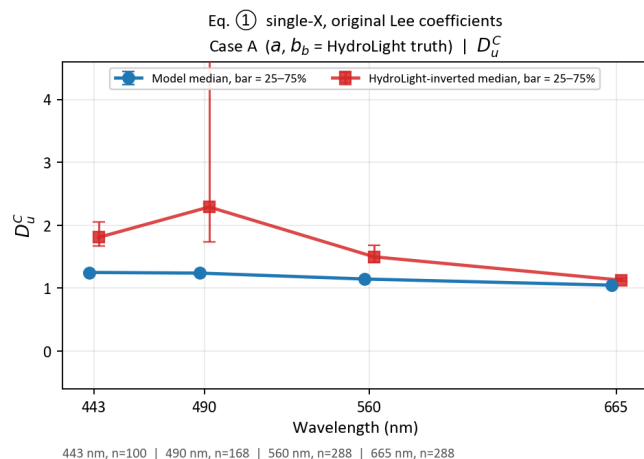


Figure A-1. Eq. ① single-X, original Lee coefficients, Case A (a, b_b = HydroLight truth): DuC spectrum. Wavelength (nm) on the x-axis; DuC (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; $n < 30$ is marked omitted.

Reading (Eq. ① (single-X, Lee)). Original Lee DuC (blue) is nearly flat at 1.05–1.25 across wavelength. HydroLight-inverted values (red) have medians 1.81 (443 nm) $\rightarrow$ 1.13 (665 nm), larger than blue and more dispersed. n is small at 443 and 490 nm (weak bottom signal at short wavelengths); treat those bands as indicative only. A single X must fit water column and bottom together, so the required Du becomes large. In Case B (right), a and b_b come from QAA, so κ and u change and both the location and the spread of the inverted values change with them.

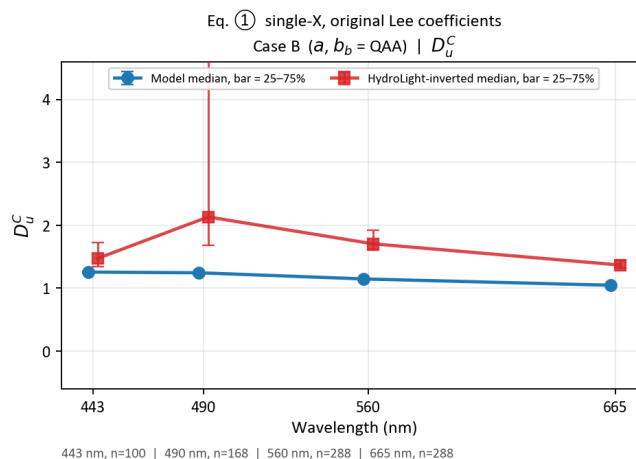


Figure A-2. Eq. ① single-X, original Lee coefficients, Case B (a, b_b = QAA): DuC spectrum. Wavelength (nm) on the x-axis; DuC (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; $n < 30$ is marked omitted.

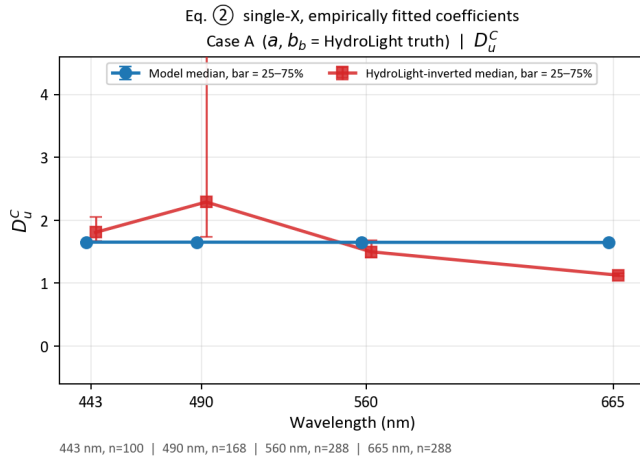


Figure A-3. Eq. ② single-X, empirically fitted coefficients, Case A ($a, b_b = \text{HydroLight truth}$): DuC spectrum. Wavelength (nm) on the x-axis; DuC (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; n < 30 is marked omitted.

Reading (Eq. ② (single-X, empirical)). Empirical DuC (blue) is nearly a constant 1.65. Inverted medians 1.50 (560 nm) and 1.13 (665 nm) lie closer to it than to Eq. ①. That is why the Rrs bias of Eq. ② is near zero. In Case B (right), a and b_b come from QAA, so κ and u change and both the location and the spread of the inverted values change with them.

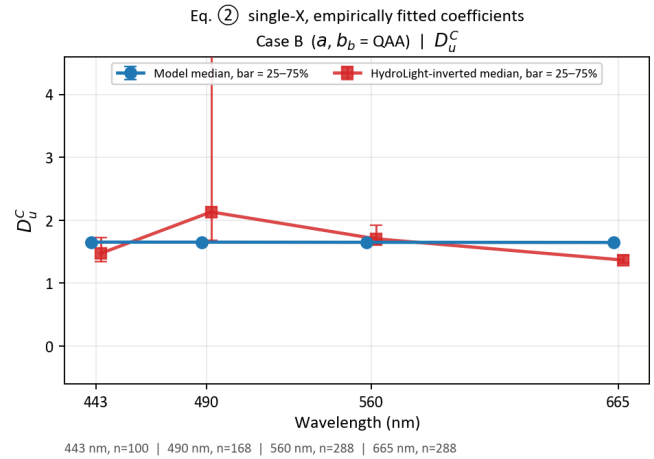


Figure A-4. Eq. ② single-X, empirically fitted coefficients, Case B ($a, b_b = \text{QAA}$): DuC spectrum. Wavelength (nm) on the x-axis; DuC (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; n < 30 is marked omitted.

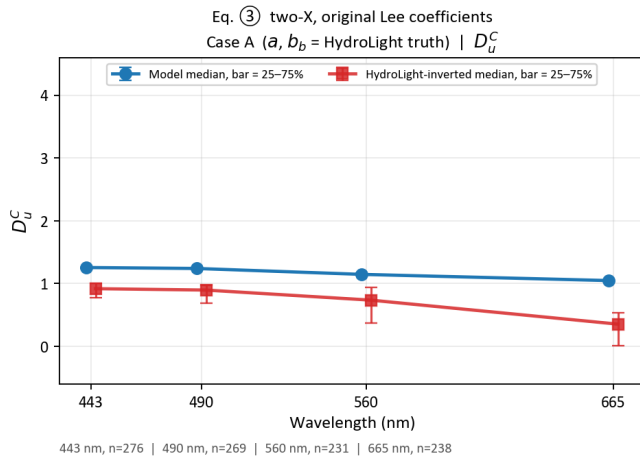


Figure A-5. Eq. ③ two-X, original Lee coefficients, Case A ($a, b_b = \text{HydroLight truth}$): DuC spectrum. Wavelength (nm) on the x-axis; DuC (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; n < 30 is marked omitted.

Reading (Eq. ③ (two-X, Lee)). In the two-X form, inverted water-column DuC (red) has medians 0.92 (443 nm) → 0.35 (665 nm), smaller than the Lee values 1.25–1.05 (blue). Original Lee coefficients overstate water-column attenuation, so modelled Rrs is too bright (positive bias of Eq. ③). In Case B (right), a and b_b come from QAA, so κ and u change and both the location and the spread of the inverted values change with them.

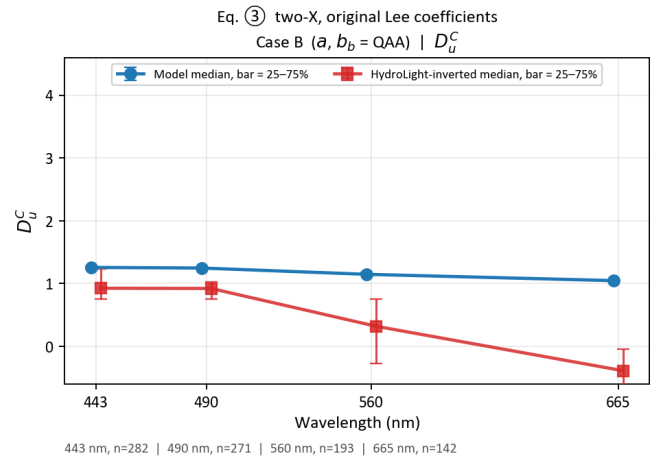


Figure A-6. Eq. ③ two-X, original Lee coefficients, Case B ($a, b_b = \text{QAA}$): DuC spectrum. Wavelength (nm) on the x-axis; DuC (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; n < 30 is marked omitted.

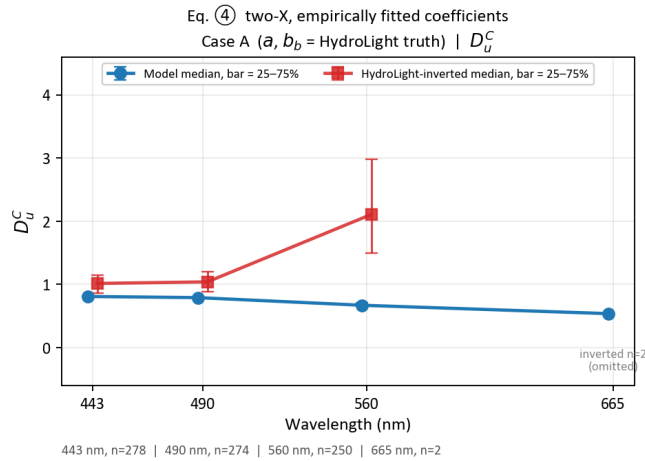


Figure A-7. Eq. ④ two-X, empirically fitted coefficients, Case A (a, b_b = HydroLight truth): DuC spectrum. Wavelength (nm) on the x-axis; DuC (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; n < 30 is marked omitted.

Reading (Eq. ④ (two-X, empirical)). Empirical DuC (blue) is low, 0.81 (443 nm) → 0.53 (665 nm). Inverted medians (red) are 1.02 (443 nm), 1.04 (490 nm), 2.11 (560 nm): within 0.2 of blue at 443–490 nm and farther at 560 nm. 665 nm has only 2 successful inversions and is omitted (the bottom term is a large constant 1.613, leaving almost no room to fit the water-column term). In Case B (right), a and b_b come from QAA, so κ and u change and both the location and the spread of the inverted values change with them.

DuC (water-column term) — band-wise RMSE

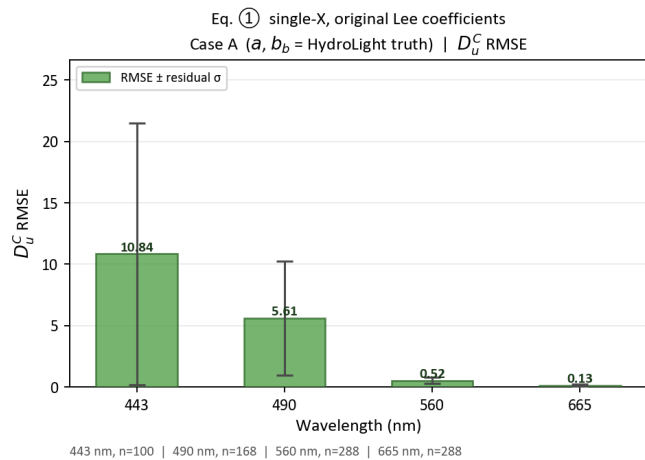


Figure A-9. Eq. ① single-X, original Lee coefficients, Case A (a, b_b = HydroLight truth): band-wise DuC RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm\sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

Reading (Eq. ① (single-X, Lee)). Case A RMSE from 443 to 665 nm: 10.84, 5.61, 0.52, 0.13; Case B: 9.23, 5.45, 0.73, 0.35. Larger values mean the formula D_u is farther from the HydroLight requirement at that band. Bands with fewer than 30 successful inversions are omitted from the figures and tables.

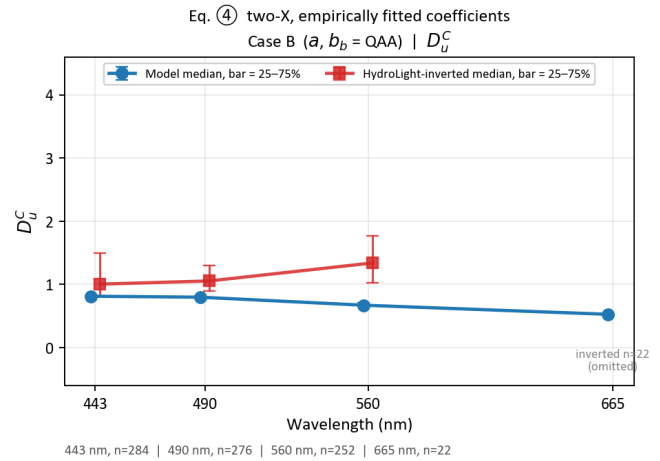


Figure A-8. Eq. ④ two-X, empirically fitted coefficients, Case B (a, b_b = QAA): DuC spectrum. Wavelength (nm) on the x-axis; DuC (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; n < 30 is marked omitted.

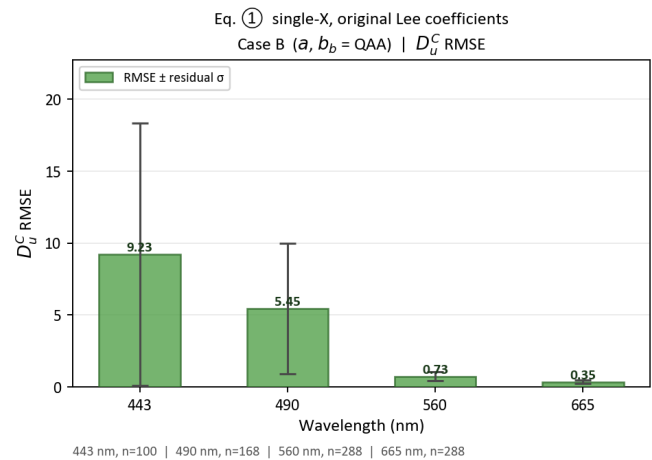


Figure A-10. Eq. ① single-X, original Lee coefficients, Case B (a, b_b = QAA): band-wise DuC RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm\sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

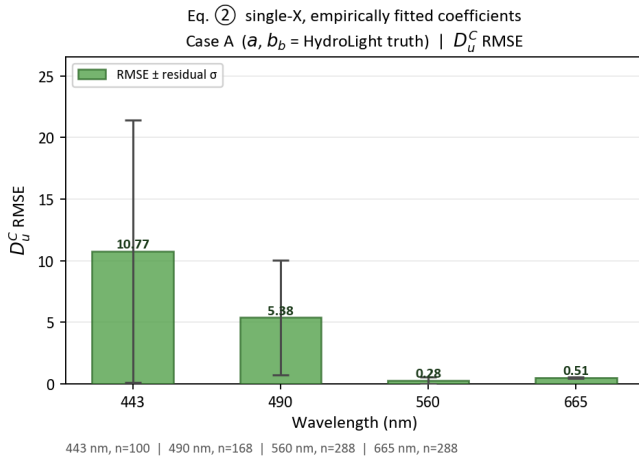


Figure A-11. Eq. ② single-X, empirically fitted coefficients, Case A ($a, b_b = \text{HydroLight truth}$): band-wise DuC RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm \sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

Reading (Eq. ② (single-X, empirical)). Case A RMSE from 443 to 665 nm: 10.77, 5.38, 0.28, 0.51; Case B: 9.17, 5.23, 0.35, 0.29. Larger values mean the formula D_u is farther from the HydroLight requirement at that band. Bands with fewer than 30 successful inversions are omitted from the figures and tables.

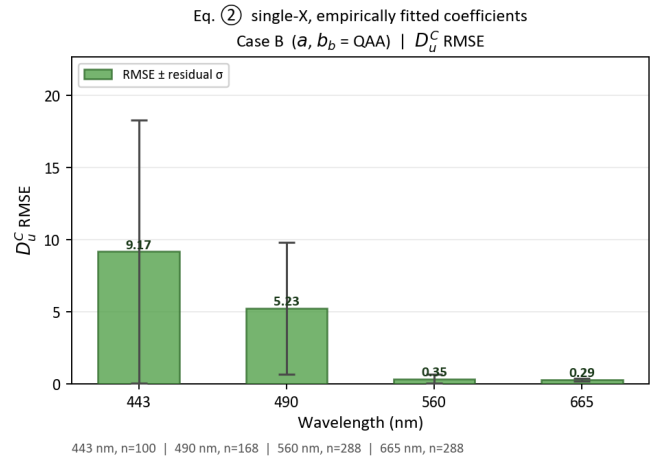


Figure A-12. Eq. ② single-X, empirically fitted coefficients, Case B ($a, b_b = \text{QAA}$): band-wise DuC RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm \sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

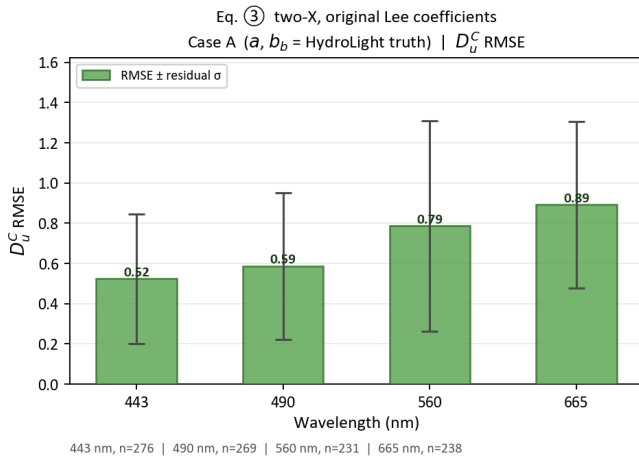


Figure A-13. Eq. ③ two-X, original Lee coefficients, Case A ($a, b_b = \text{HydroLight truth}$): band-wise DuC RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm \sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

Reading (Eq. ③ (two-X, Lee)). Case A RMSE from 443 to 665 nm: 0.52, 0.59, 0.79, 0.89; Case B: 0.67, 0.51, 1.11, 1.47. Larger values mean the formula D_u is farther from the HydroLight requirement at that band. Bands with fewer than 30 successful inversions are omitted from the figures and tables.

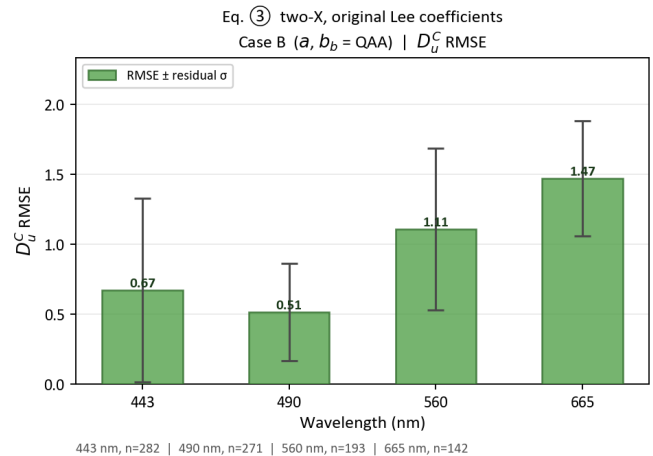


Figure A-14. Eq. ③ two-X, original Lee coefficients, Case B ($a, b_b = \text{QAA}$): band-wise DuC RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm \sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

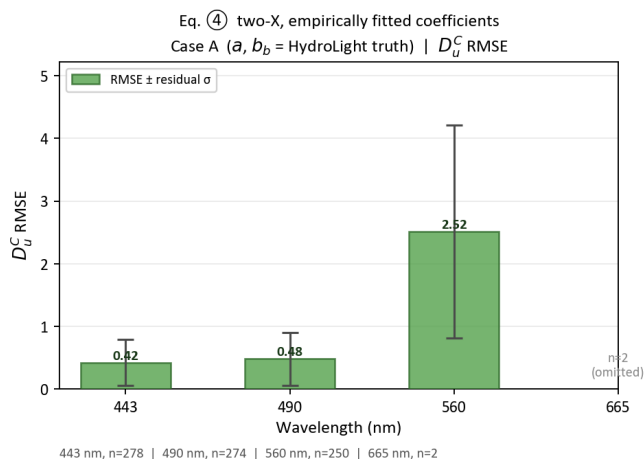


Figure A-15. Eq. ④ two-X, empirically fitted coefficients, Case A (a, b_b = HydroLight truth): band-wise DuC RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm\sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

Reading (Eq. ④ (two-X, empirical)). Case A RMSE from 443 to 665 nm: 0.42, 0.48, 2.52, —; Case B: 1.06, 0.63, 1.28, —. Larger values mean the formula Du is farther from the HydroLight requirement at that band. Bands with fewer than 30 successful inversions are omitted from the figures and tables.

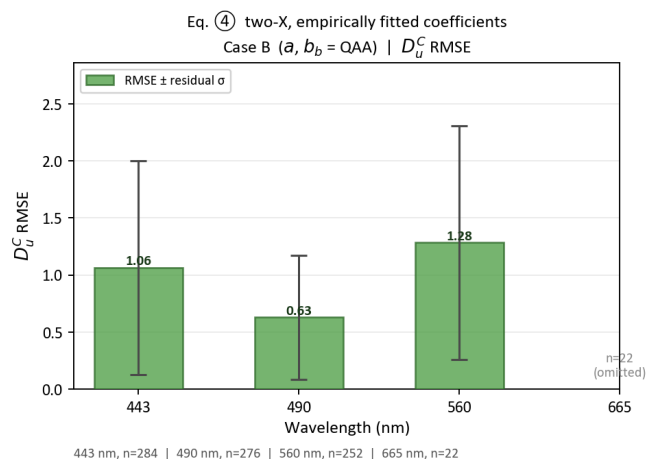


Figure A-16. Eq. ④ two-X, empirically fitted coefficients, Case B (a, b_b = QAA): band-wise DuC RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm\sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

Eq.	Case A (HydroLight a, b_b)				Case B (QAA a, b_b)			
RMSE	443	490	560	665	443	490	560	665
①	10.84	5.61	0.52	0.13	9.23	5.45	0.73	0.35
②	10.77	5.38	0.28	0.51	9.17	5.23	0.35	0.29
③	0.52	0.59	0.79	0.89	0.67	0.51	1.11	1.47
④	0.42	0.48	2.52	— (n=2)	1.06	0.63	1.28	— (n=22)

Table A-1. Band-wise DuC RMSE (formula – HydroLight-inverted; dimensionless). Matches the bar plots above. Orange background = Case B.

DuB (bottom term; two-X Eqs. ③ and ④ only) — spectra

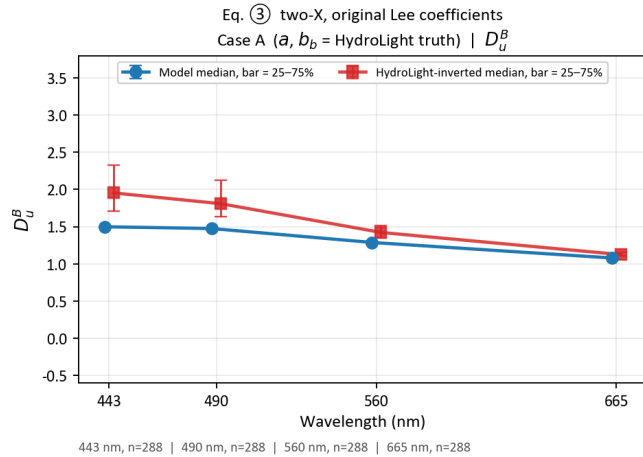


Figure A-17. Eq. ③ two-X, original Lee coefficients, Case A (a, b_b = HydroLight truth): DuB spectrum. Wavelength (nm) on the x-axis; DuB (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; $n < 30$ is marked omitted.

Reading (Eq. ③ (two-X, Lee)). Bottom term DuB: original Lee $1.04\sqrt{1+5.4u}$ (blue) is 1.50 (443 nm) → 1.08 (665 nm); inverted medians (red) are 1.95 (443 nm) → 1.12 (665 nm). At short wavelengths HydroLight requires stronger bottom attenuation than Lee. $n = 288$ at all four bands. In Case B (right), a and b_b come from QAA, so κ and u change and both the location and the spread of the inverted values change with them.

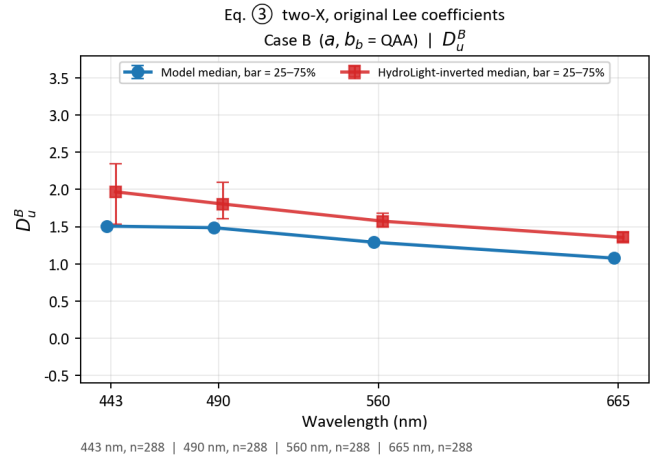


Figure A-18. Eq. ③ two-X, original Lee coefficients, Case B (a, b_b = QAA): DuB spectrum. Wavelength (nm) on the x-axis; DuB (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; $n < 30$ is marked omitted.

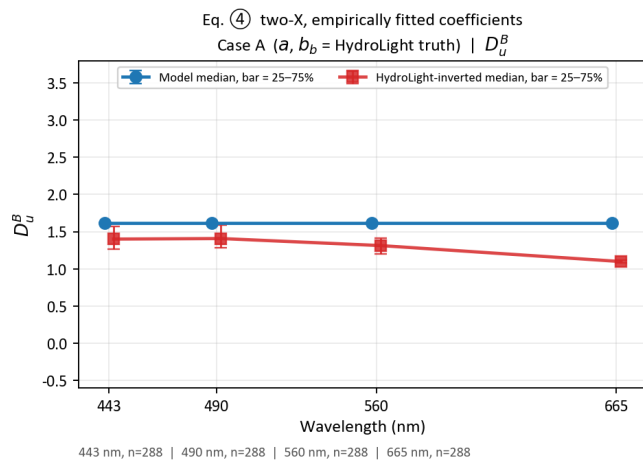


Figure A-19. Eq. ④ two-X, empirically fitted coefficients, Case A (a, b_b = HydroLight truth): DuB spectrum. Wavelength (nm) on the x-axis; DuB (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; $n < 30$ is marked omitted.

Reading (Eq. ④ (two-X, empirical)). Empirical DuB = 1.613 constant (blue). Inverted medians (red) are 1.40 (443 nm) → 1.10 (665 nm); blue is 0.2–0.5 larger. A slightly large bottom term and a small water-column term is the combination that matches Rrs. In Case B (right), a and b_b come from QAA, so κ and u change and both the location and the spread of the inverted values change with them.

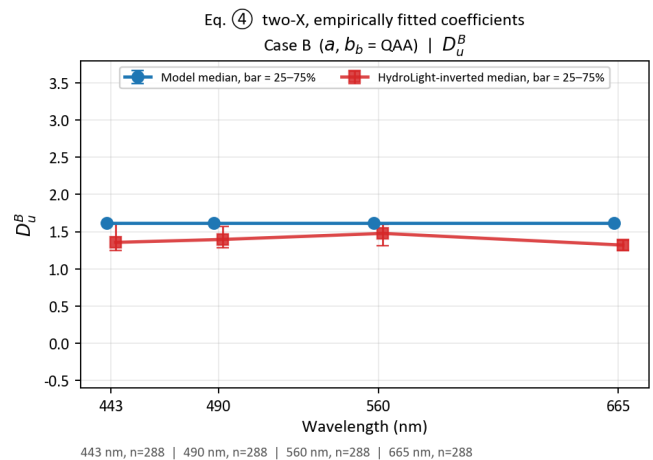


Figure A-20. Eq. ④ two-X, empirically fitted coefficients, Case B (a, b_b = QAA): DuB spectrum. Wavelength (nm) on the x-axis; DuB (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; $n < 30$ is marked omitted.

DuB (bottom term; two-X Eqs. ③ and ④ only) — band-wise RMSE

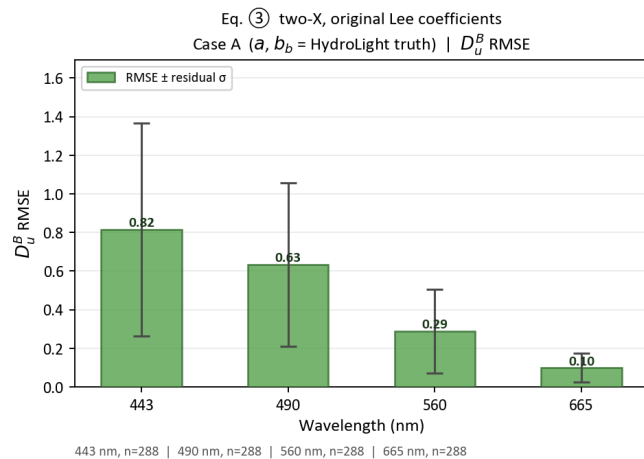


Figure A-21. Eq. ③ two-X, original Lee coefficients, Case A (a, b_b = HydroLight truth): band-wise DuB RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm\sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

Reading (Eq. ③ (two-X, Lee)). Case A RMSE from 443 to 665 nm: 0.82, 0.63, 0.29, 0.10; Case B: 1.04, 0.63, 0.43, 0.31. Larger values mean the formula D_u is farther from the HydroLight requirement at that band. Bands with fewer than 30 successful inversions are omitted from the figures and tables.

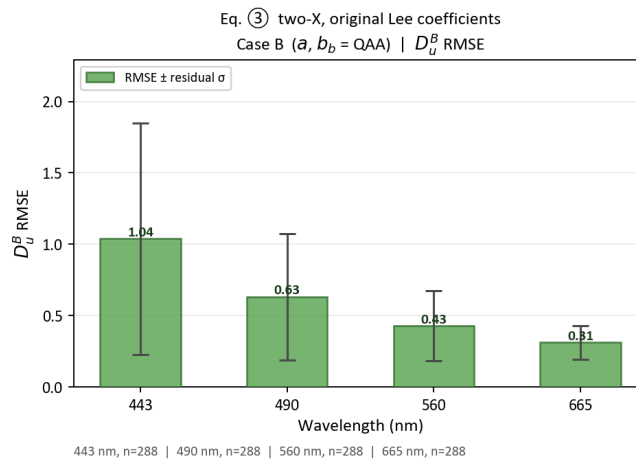


Figure A-22. Eq. ③ two-X, original Lee coefficients, Case B (a, b_b = QAA): band-wise DuB RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm\sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

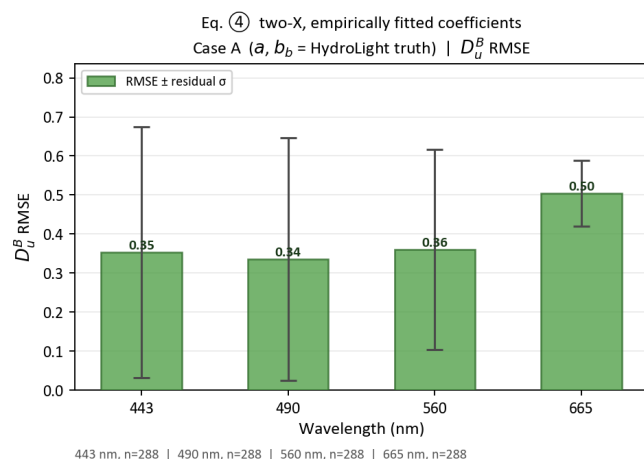


Figure A-23. Eq. ④ two-X, empirically fitted coefficients, Case A (a, b_b = HydroLight truth): band-wise DuB RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm\sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

Reading (Eq. ④ (two-X, empirical)). Case A RMSE from 443 to 665 nm: 0.35, 0.34, 0.36, 0.50; Case B: 0.37, 0.33, 0.31, 0.31. Larger values mean the formula D_u is farther from the HydroLight requirement at that band. Bands with fewer than 30 successful inversions are omitted from the figures and tables.

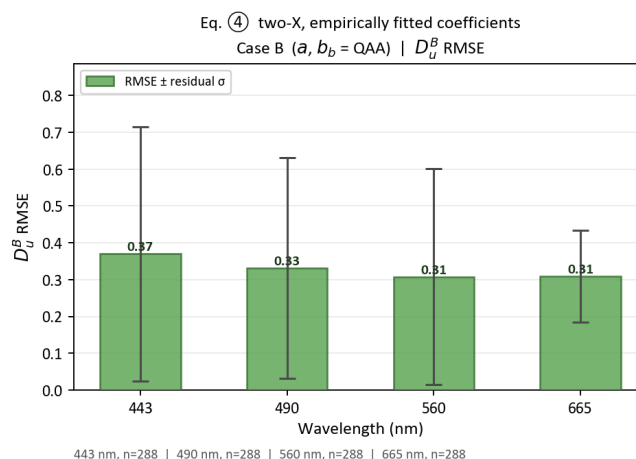


Figure A-24. Eq. ④ two-X, empirically fitted coefficients, Case B (a, b_b = QAA): band-wise DuB RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm\sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

Eq.	Case A (HydroLight a, b_b)				Case B (QAA a, b_b)			
RMSE	443	490	560	665	443	490	560	665
③	0.82	0.63	0.29	0.10	1.04	0.63	0.43	0.31
④	0.35	0.34	0.36	0.50	0.37	0.33	0.31	0.31

Table A-2. Band-wise DuB RMSE (formula – HydroLight-inverted; dimensionless). Matches the bar plots above. Orange background = Case B.